

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

That's how we validate data!



Learning Outcome:

Designs the data structure based on the database structures and its characteristics.

Validate database using the proper IT tools.

Can Do Statements:

- I can talk about the different data validation methods.
- I can describe the different measured parameters checks.
- I can tell what the benefits of performing data validation are.

Ask your partner:

- According to you, how hard is it to validate data? Why?
- Do you think that data can be fully validated by machines in the near future? Why? Why not?





Let's Explore!

Getting Ready for a New Job



Instructions: Read the text below.

Mariana: Dan, you used to work for TeCHno industries, right?

Dan: Yes, I used to be one of the professionals that carried out validation processes on the different company's systems.

Mariana: I am starting my new job this week. What can you tell me about data validation?

Dan: I can remember that there used to be two parts to any validation method. The first one, the data screening, would implement a series of different algorithms and codes to screen all the data values that could be doubtful, resulting in a report of the data validity. The second one, the data verification, would retake the suspicious values and decide what the proper action would be for each one. In other words, a professional will decide which and how to address the values by validating, rejecting, replacing or fixing them.

Mariana: Great, what about the measured parameters check?

Dan: It is one of the validation routines that will measure, through three different tests, various parameters before the values are incorporated in the data base. The range test is going to compare the data with pre-established limits and assess it in different criteria. The relational test is going to corroborate that improbable situations of the real world are not included in the data without proper verification. Finally, the trend test is going to analyze the changes that data, and some criteria, suffer through an established period of time.

Mariana: Why did the data validation use to be so important?

Dan: Well, it used to be and still is really important. Its main objective is to detect and fix, on the shortest notice, erroneous data as well as to identify their cause in order to ensure the proper operation of applications, and avoid any similar flaws in future projects.



**Let's Practice!**

Before talking about data validation, you must take into account these concepts.

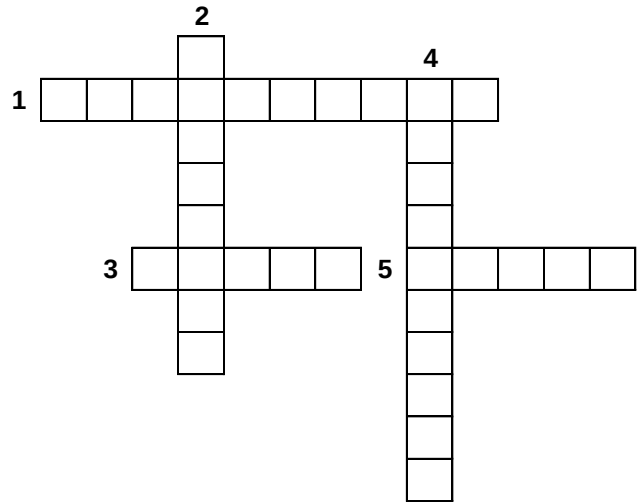
Instructions: Find the words in the word search below.

Down

2. We are going to measure the data through three different tests before we incorporate those values in the _____.
4. The _____ test is going to corroborate that improbable situations of the real world are not included in the data without proper verification.

Across

1. We will learn that the measure _____ check is a validation routine.
3. The _____ test is going to compare data with pre-established limits and assess it in different criteria.
5. Finally, the _____ test is going to analyze the changes that data, and some criteria, suffer through an established period of time.

**Let's Practice!****Data Validation Some Years Ago and Its Importance Nowadays**

Instructions: Read the sentences. Write the missing word(s) in the blank. Use: used to avoid, used to only identify, used to think, used to be, used to wait.

1. I _____ that data validation was not important.
2. The main objective of data validation _____ to detect erroneous data, but now it fixes it too.
3. Data validation _____ the error in the data information, but now it is in charge of determining its causes.
4. Software developers _____ any similar flaw using more than 200 lines of code.
5. Software developers _____ until the implementation of data to obtain the errors that databases could present, but now they apply data validation processes first.





Extension Activity

Let's plan how to lead the validation process.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about the validation methods and the steps to apply a data validation process.
- Choose a validation method and make a poster in which you explain its main features and steps.
- Share with your partner and class your poster.

Self-Assessment

Yes

No

- I look for more information about the validation methods and the steps I have to follow to apply them.
- I choose a validation process and make a poster where I explain its main features and steps.
- I share my poster with my partners and class.

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Target Words

- **criteria** (noun): plural form of "criterion"; a series of standards
e.g. The range test is going to compare the data with pre-established limits and assess it in different criteria.
- **through** (preposition): with the implementation of
e.g. It is one of the validation routines that will measure, through three different tests, various parameters before the values are incorporated in the data base.
- **on the shortest notice** (expression): as fast as possible, with the least delay
e.g. Its main objective is to detect and fix, on the shortest notice, erroneous data as well as to identify their cause in order to ensure the proper operation of applications and avoid any similar flaws in future projects.

- **screen** (verb): to project or show something that was not at sight
e.g. The first one, the data screening, would implement a series of different algorithms and codes to screen all the data values that could be doubtful, resulting in a report of the data validity.
- **doubtful** (adjective): questionable, unlikely to be successful or accurate
e.g. The first one, the data screening, would implement a series of different algorithms and codes to screen all the data values that could be doubtful, resulting in a report of the data validity.
- **still** (adverb): that is not finished yet
e.g. Well, it used to be and still is really important.



Learning Points

Used to

You use the structure **used to** to talk about past habits.

We use the expression **used to** to introduce the idea that someone had a habit, a recurrent action in the past, which is no longer happening in the present time. It can also be used to describe the reality in the past, without making reference to a specific action of a person.

The structure for creating sentences with this expression is the following:

Subject + **used to** + **verb (base form)** + complement.

For example:

*We **used to use** this software programming system 5 years ago.* (Regular action in the past)

*That system **used to be** the best one available in the market.* (Description of the past reality)

Examples

- *I used to be one of the professionals that carried out validation processes on the different company's systems.*
- *I can remember that there used to be two parts to any validation method.*





Learning Points

Measured Parameters Checks in Data Validation

The process that takes raw data and transforms it into validated data must follow certain methods.

Different tests are applied to the data in order to validate it. Check the following chart to learn more about these tests:

MEASURED PARAMETER CHECK	DESCRIPTION
Range Test	This validation test compares the measured data with pre-established limits and assesses it through different types of criteria. The obtained values can be either positive or negative, reflecting the status of the analyzed data. Each criterion whose value fits within the limits is considered valid.
Relational Test	The relational test corroborates that improbable situations of the real world are not stated in the data without proper verification.
Trend Test	The trend test assesses the changes present in different values and criteria in a specific period of time.

